Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 4 · SLO 2 — Solved Worksheet

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Unit â€“ 3 : Session â€“ 4 : SLO â€“ 2 | Stack Applications â€” Infix to Postfix Conversion

I. Algorithm: Infix to Postfix Using a Stack

• Step 1: Scan the infix expression from left to right.
• Step 2: If the scanned symbol is an OPERAND, add it to the postfix expression.
• Step 3: If the scanned symbol is '(', PUSH it onto the stack.
• Step 4: If the scanned symbol is ')', POP and output all symbols until '(' is found; discard '('.
• Step 5: If the scanned symbol is an OPERATOR:
• a) If the stack is empty or top is '(', push the operator.
• b) Else, pop all operators from the stack whose precedence is GREATER THAN OR EQUAL to
• the scanned operator, output them, then push the scanned operator.
• c) Note: '^' is right-associative â€” pop only operators of strictly higher precedence.
• Step 6: Repeat Steps 1-5 until the entire infix expression is scanned.
• Step 7: Pop and output all remaining operators from the stack.
• Step 8: EXIT

II. Example: A + B * C -> ABC*+

• Scan A -> output A. Stack: empty
• Scan + -> push. Stack: +
• Scan B -> output B. Output: AB
• Scan * -> * has higher precedence than + -> push. Stack: + *
• Scan C -> output C. Output: ABC
• End -> pop * and +. Output: ABC*+